Searching Legislative Design Space for Compromise and Productivity

A Computational Framework for Comparing Congress-Style Institutional Designs

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Modern legislatures are often judged by two goals that pull against each other: they should be productive enough to act, and they should be representative enough to convert competing public interests into legitimate compromise. This paper presents a computational framework for searching a broad legislative design space rather than defending one constitutional reform. The simulator compares Congress-style institutional systems under shared assumptions, including ordinary majority rule, supermajority rule, default enactment unless a blocking coalition forms, challenge vouchers, cross-bloc cosponsorship, adaptive procedural tracks, sunset review, and earned proposal credits.

The framework separates productivity from compromise, public responsiveness, agenda control, and legitimacy-related risk. Legislators are modeled as individual actors with heterogeneous ideal points and voting sensitivities; bills are proposed around agenda setters; enacted bills update a scalar status quo; and institutional procedures are implemented as composable stages. Across 18 assumption cases and 19 institutional scenarios in the current campaign, open default enactment produces high throughput but also high low-support passage, larger policy shifts, and greater proposer gain. Committee guardrails sharply reduce those risks at large productivity cost. Explicit budgeted lobbying makes organized-interest spending and defensive lobbying against anti-lobbying reform measurable. Challenge-voucher and tokenless escalation sweeps reveal a throughput/safety frontier. Cross-bloc cosponsorship gates improve welfare and reduce proposer advantage by requiring coalition breadth before agenda access, but can become blunt screens when applied uniformly. Adaptive procedural tracks preserve more agenda access while routing risky bills into stronger review lanes. Sunset trial review makes risky enactments reversible, improving welfare and reducing policy volatility without a front-end agenda gate. Earned proposal credits add mild stateful proposer accountability, reducing floor load and policy shift more gently than committee or cross-bloc gates. Structured mediation moves many default-pass bills toward the chamber median and status quo, reducing policy shift and proposer gain while preserving high throughput. The contribution is a reproducible mechanism-search framework for identifying institutional designs that can improve compromise and productivity while remaining responsive to public support and generated public benefit.

CCS Concepts: • **Human-centered computing → Collaborative and social computing theory, concepts and paradigms**; • **Computing methodologies → Modeling and simulation**; • **Applied computing** → *Law*.

Additional Key Words and Phrases: legislative institutions, institutional design, computational simulation, agenda control, compromise, collective decision-making, voting systems

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1 Introduction

The motivating question is not whether one unusual voting rule should replace Congress. It is whether a systematic design search can discover legislative structures that better combine productivity, compromise, and public responsiveness. A legislature can be highly unified without being democratic; authoritarian systems can appear cohesive because dissent has been suppressed. A useful legislative system should instead help representatives with competing interests reach

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compromises that are productive, broadly supportable, and connected to the preferences and welfare of the people they represent.

That goal makes raw legislative productivity a poor target by itself. A system that passes many laws may be efficient, but it may also be volatile, captured, or weakly legitimate. A strict consensus system may be cooperative but unable to adapt. A simple majority system may be responsive to a median coalition but inattentive to concentrated minority harm. The core problem is therefore comparative: which institutional designs produce the best tradeoffs among productivity, compromise, representative responsiveness, minority protection, and resistance to agenda capture?

We build an individual-legislator simulation framework to study that question. The simulator compares institutional rules under shared generated worlds and reports a dashboard of outcomes: enacted bills, floor load, welfare proxy, coalition support, low-support passage, policy movement, proposer gain, gridlock, committee rejection, and challenge use. Each final choice is still a yes-or-no vote, but bills may pass through proposal access, committee information, committee gatekeeping, challenge windows, adaptive routing, sunset review, bicameral approval, or executive veto before that vote.

Default enactment is one test case within that broader search. It is useful because it sharply reverses the burden of collective action: proposed laws pass unless a blocking coalition stops them. But the simulator is not designed to validate default enactment as the preferred reform. It is designed to compare default enactment against ordinary thresholds, coalition-breadth gates, scarce challenge rights, proposer accountability, reversible lawmaking, and future systems that may be more radically different. The central thesis is that legislative design should be evaluated as a mechanism portfolio: agenda rights, proposal volume, screening costs, information institutions, coalition-breadth requirements, blocking rights, reversibility, and proposer accountability jointly determine whether a system produces compromise or merely produces output.

2 Related Work

The project sits between computational political modeling, formal legislative theory, and institutional design. Individual-level simulation models are useful when heterogeneous actors interact under rules that create path-dependent aggregate outcomes [de Marchi and Page 2014; Epstein 2006; Gilbert and Troitzsch 2005]. Prior work has used such models to explore political representation and legislative performance, including simulations of parliament composition and decision efficiency [Pluchino et al. 2011]. We use the method differently: the object of comparison is not the demographic composition of a legislature, but the procedural architecture through which proposals reach enactment.

Formal legislative theory motivates the model structure. Spatial voting motivates one-dimensional policy positions and status-quo-relative utility [Downs 1957]. Legislative bargaining and agenda-setter models motivate explicit proposer identity and proposer gain [Baron and Ferejohn 1989]. Veto-player theory and pivotal-politics models motivate comparing simple majorities, supermajorities, bicameralism, and executive vetoes [Krehbiel 1998; Tsebelis 2002]. Agenda-control theories motivate treating bills that never reach the floor as substantively important, not as missing observations [Cox and McCubbins 2005]. Empirical legislative-effectiveness work further warns that productivity and influence should not be collapsed into a single measure [Volden and Wiseman 2014].

One tested design family, default enactment, has real institutional analogues, although not usually in broad ordinary lawmaking. WTO dispute-settlement reports are adopted unless there is appeal or consensus not to adopt [World Trade Organization 1994]. EU fiscal-governance procedures use reverse qualified-majority mechanisms in some enforcement contexts [European Parliament and Council of the European Union 2011]. Tacit-acceptance treaty amendment procedures and silence procedures allow decisions to take effect unless objections are raised by a deadline [Churchill and Ulfstein

2000]. In the United States, review-and-disapproval structures such as the Congressional Review Act let agency rules take effect unless Congress and the President complete a disapproval process [Carey and Davis 2024; United States Congress 1996].

These analogues share a broader design lesson. Institutional rules rarely operate alone. Burden-shifted decision rules tend to be paired with constrained proposers, bounded subject matter, information review, objection windows, or post-enactment correction. Likewise, ordinary majority rules, committee systems, and supermajority rules behave differently depending on agenda access and bargaining opportunities. The simulator therefore treats each scenario as a bundle of mechanisms, not as a single isolated threshold.

3 Model

Each simulation run creates a legislature, a stream of bills, a scalar status quo, and an institutional process. The model is intentionally austere so that procedural effects are visible.

3.1 Actors and Bills

Legislators have ideal points in $[-1, 1]$, party labels, and heterogeneous sensitivities to party, constituency, lobbying, reputation, and compromise. A legislator's vote is generated by a weighted strategy that evaluates the bill relative to the current status quo and to the legislator's own incentive profile.

Bills have a proposer, a policy position in $[-1, 1]$, public support, public-benefit score, lobby pressure, and salience. Proposals are generated around proposers so that agenda-setter advantage is measurable. When a bill is enacted, the status quo moves to that bill's policy position. Votes are therefore not abstract preferences over isolated proposals; they are evaluations of a policy change.

3.2 Institutional Stages

The implementation separates voting rules from legislative processes. A voting rule determines how a chamber interprets yes and no votes. A legislative process determines what stages a bill must pass through. Current processes include unicameral majority rule, 60 percent supermajority passage, default passage unless two-thirds block, bicameral majority rule, presidential veto with override, proposal-access screens, proposal-cost access, cross-bloc cosponsorship access, committee gatekeeping, committee information review, challenge vouchers, adaptive routing, sunset review, earned proposal credits, budgeted lobbying, and structured amendment mediation.

This modular design matters because no legislative rule is meaningful in isolation. The same final threshold can behave differently when proposals are open to all legislators, filtered by a committee, priced by agenda costs, routed by risk, exposed to scarce challenge rights, or revised through amendment. The framework is intended to make those combinations easy to test.

3.3 Challenge Vouchers

In the challenge-voucher process, bills start in a default-enactment lane. Parties receive a limited number of challenge tokens per session. Legislators evaluate whether a bill is sufficiently harmful or risky to justify spending a party token. If a challenge is filed, the bill leaves the default lane and is routed to an active affirmative vote. If no challenge is filed, the bill is enacted by default.

The voucher mechanism tests whether default enactment can preserve fast treatment for low-conflict bills while forcing contested bills into a higher-legitimacy lane. It differs from ordinary veto power because objection capacity is scarce. It differs from committee gatekeeping because opponents must actively spend a limited procedural resource.

3.4 Cross-Bloc Cosponsorship

The cross-bloc cosponsorship process sits before floor consideration. A proposer must attract credible support from legislators outside the proposer's party bloc and at a minimum ideological distance. The rule estimates whether potential outside-bloc cosponsors would publicly attach themselves to the bill based on policy fit, public support, generated public benefit, compromise preference, status-quo shift, and lobby-risk signals.

This mechanism tests whether agenda access can reward coalition breadth without relying on a permanent committee veto. It is intentionally weaker than a final vote: a bill still ultimately passes or fails under the downstream institutional process. The gate only asks whether the proposal has enough cross-bloc credibility to deserve the default lane.

3.5 Adaptive Tracks

Adaptive tracks route each bill to a procedural lane according to a generated risk score. The score combines low public support, policy shift, positive lobby pressure, salience, disagreement between public support and generated public benefit, and low generated benefit. Low-risk bills stay in a fast default-pass lane. Middle-risk bills go either to active simple-majority review or to challenge-voucher review, depending on the scenario. High-risk bills go to informed guardrails.

This mechanism tests whether routine and risky bills should receive different procedures rather than being forced through one uniform legislative track. It also exposes a design choice: the middle lane can prioritize stronger consent, or it can prioritize throughput by using scarce challenge rights.

3.6 Sunset Trial Review

Sunset trial review treats risky enacted bills as provisional. After the downstream process enacts a bill, the sunset wrapper applies an automatic review based on generated public benefit, public support, lobby risk, and the same risk score used by adaptive tracks. If the review score is high enough, the law is renewed and the status quo remains at the enacted policy. If review fails, the trial expires and the status quo rolls back.

The current implementation is an abstract review, not a multi-period law registry. It is still useful because it isolates reversibility from front-end gatekeeping: the bill can enter and pass through the default lane, but weak post-enactment evidence prevents permanent policy movement.

3.7 Earned Proposal Credits

Earned proposal credits replace a static proposal-cost screen with stateful agenda rights. Each proposer starts with a small credit balance and receives a modest session drip. Proposals spend credits according to generated risk, positive lobby pressure, salience, and public value. After the downstream process considers the bill, the proposer earns or loses credits based on public benefit, public support, risk, lobby pressure, and whether the bill passed with weak support.

This mechanism tests whether agenda access can learn from proposer track records. It differs from a fixed proposal cost because bad proposals reduce future access for that proposer, while high-quality proposals replenish agenda capacity. It is still intentionally simple: credits are attached to individual proposers, not parties or coalitions, and the current metric suite does not yet report proposer concentration directly.

3.8 Budgeted Lobbying

Budgeted lobbying makes organized interests explicit actors rather than only bill-level pressure. Lobby groups have issue domains, preferred positions, budgets, influence intensity, information bias, public-campaign skill, and defensive multipliers. A budgeted lobbying wrapper lets groups spend on bills that align with their preferred position and spend defensively against anti-lobbying reform. The mechanism is not intended to validate lobbying as desirable. It is included so capture pressure, defensive spending, and public-preference distortion can be measured rather than folded into a single voting weight.

3.9 Structured Amendment Mediation

Structured mediation adds a bounded content-change stage before the final yes-or-no decision. When a bill has weak public support, large policy shift, high capture risk, or strong proposer advantage, the mediator moves the bill toward a weighted target combining the chamber median, the current status quo, and the proposer position. The amended bill then proceeds through the downstream institutional process. This does not model full legislative bargaining, but it adds an essential distinction: a system can improve compromise by revising proposal content, not only by passing, blocking, or reviewing fixed proposals.

4 Metrics

The simulator reports a metric suite rather than a single score:

- Productivity: share of potential bills enacted.
- Floor consideration: share of potential bills reaching the floor.
- Enacted bills per run and floor bills per run: absolute throughput and institutional load.
- Welfare: average generated public-benefit score of enacted bills.
- Cooperation: productivity weighted by enacted support.
- Compromise: support, moderation, and distance from proposer advantage.
- Low-support passage: enacted bills with less than 50 percent yes support.
- Popular failure: high-public-support bills that fail.
- Policy shift: average absolute movement from the status quo.
- Proposer gain: enacted movement toward the proposer's ideal point.
- Challenge rate: share of potential bills diverted from default enactment into active voting.
- Amendment rate and amendment movement: share of potential bills revised by mediation, and average absolute movement before voting.
- Lobby spend and defensive spend: explicit organized-interest spending per potential bill, and the share spent against anti-lobbying reform.

The most important distinction is between productivity and compromise. A high-throughput institution may still enact bills with little active support. Conversely, a low-throughput institution may show high welfare among enacted bills but fail to adapt. The current compromise proxy is only a first step: the long-term target is to model representative compromise more directly by measuring public support, public benefit, coalition breadth, movement away from proposer advantage, and protection against concentrated minority harm.

5 Experimental Design

The current campaign, `simulation-campaign-v10`, runs 150 simulations per case, with 101 legislators, 60 base bills per run, base seed 20260428, 18 assumption cases, and 19 institutional scenarios per case. The assumption cases vary polarization, party loyalty, compromise culture, lobbying pressure, constituency pressure, party count, proposal pressure, flooding conditions, capture pressure, and public appetite for anti-lobbying reform.

The institutional scenarios are:

- unicameral simple majority;
- unicameral 60 percent passage;
- default pass unless two-thirds block;
- default pass plus scarce challenge vouchers;
- default pass plus cross-bloc cosponsorship gates;
- adaptive procedural tracks that route bills by generated risk;
- sunset trial legislation with automatic review for risky enactments;
- earned proposal credits;
- explicit budgeted lobbying, transparency, and anti-capture bundles;
- structured amendment mediation for simple-majority and default-pass systems;
- default pass plus informed guarded committee;

The proposal-cost result reported below comes from `simulation-campaign-v2`, which included proposal costs, bicameralism, and presidential veto scenarios. Challenge-token sweeps over party budgets, member tokens, and q-member escalation come from earlier campaigns. The v10 campaign integrates the main mechanism families with explicit lobbying actors and structured amendment mediation so that the headline comparison remains interpretable.

The campaign is deterministic and reproducible from the repository with `make campaign`. The paper is generated with `make paper`.

6 Code Availability

The simulator source code, LaTeX manuscript source, generated campaign reports, and tracked paper PDF are available in the public open-source repository: https://github.com/Jacoba1100254352/congress-institutional-simulator.

7 Results

Table 1 summarizes selected scenario averages across all v10 assumption cases. The table mixes one absolute load measure with normalized metrics: *Enacted/run* is a count of enacted bills per simulated run, while the other columns are rates or scores on a 0–1 scale. Figures 1 and 2 visualize the same tradeoff space.

7.1 Open Default Enactment

Open default enactment is consistently the broad throughput leader among baseline institutions. Across v10, it averages 0.842 productivity and 86.912 enacted bills per run. By comparison, unicameral simple majority averages 0.265 productivity and 26.189 enacted bills per run, while the 60 percent supermajority rule averages 0.091 productivity and 8.584 enacted bills per run.

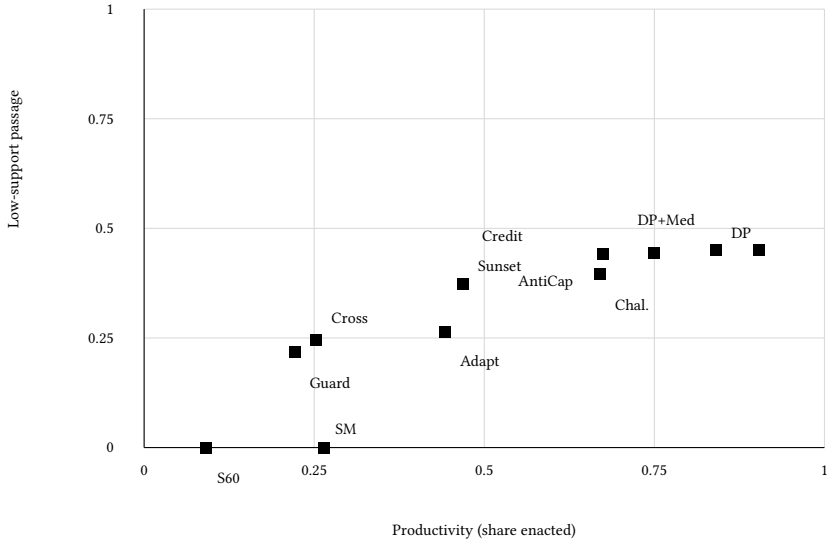

Fig. 1. Productivity versus low-support passage for selected v10 scenarios. The figure makes the main scale distinction visible: productivity and low-support passage are normalized rates, while Table 1 also reports enacted bills per run as an absolute load measure.

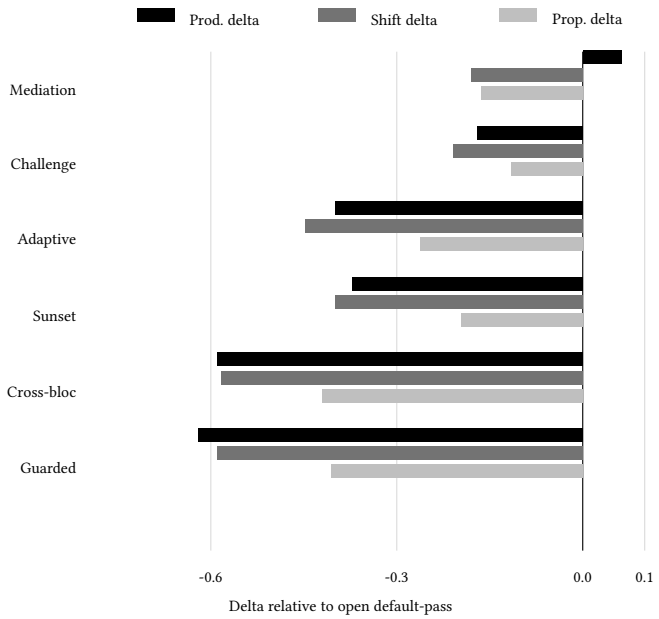

Fig. 2. Productivity, policy-shift, and proposer-gain deltas relative to open default-pass. Negative shift and proposer-gain deltas are desirable; productivity deltas show the throughput cost or gain.

Table 1. Selected scenario averages across 18 assumption cases in `simulation-campaign-v10`.

Scenario	Prod.	Enacted/run	Welfare	Low-sup.	Shift	Prop. gain	Challenge
Default pass unless 2/3 block	0.842	86.912	0.496	0.451	0.663	0.656	0.000
Default pass + mediated amendments	0.903	93.007	0.498	0.452	0.483	0.492	0.000
Default pass + budgeted lobbying	0.842	86.868	0.496	0.452	0.663	0.655	0.000
Default pass + challenge vouchers	0.670	77.968	0.509	0.394	0.455	0.539	0.398
Default pass + adaptive tracks	0.442	44.887	0.605	0.264	0.216	0.393	0.000
Default pass + sunset trial	0.469	47.595	0.618	0.374	0.265	0.460	0.000
Default pass + earned credits	0.674	64.552	0.513	0.442	0.508	0.624	0.000
Default pass + cross-bloc cosponsors	0.253	24.696	0.687	0.245	0.080	0.236	0.000
Default pass + informed guarded committee	0.221	22.067	0.651	0.217	0.073	0.250	0.000
Default pass + anti-capture bundle	0.750	76.597	0.518	0.445	0.585	0.649	0.000
Unicameral simple majority	0.265	26.189	0.637	0.000	0.102	0.307	0.000
Unicameral 60 percent passage	0.091	8.584	0.707	0.000	0.020	0.148	0.000

Note: Enacted/run is an absolute count. All other reported columns are normalized rates or scores.

The throughput gain comes with risk signals. Open default enactment averages 0.451 low-support passage, 0.663 policy shift, and 0.656 proposer gain. In this model, the rule is not merely reducing gridlock; it is also allowing many proposals to take effect without majority support and moving policy substantially toward proposers.

7.2 Guardrails

Informed guardrails sharply reduce these risks. Default pass with access screening, committee information, and committee gatekeeping averages 0.221 productivity, 22.067 enacted bills per run, 0.651 welfare, 0.217 low-support passage, 0.073 policy shift, and 0.250 proposer gain. Relative to open default enactment, that is a 0.234 reduction in low-support passage and a 0.590 reduction in policy shift, but also a 0.621 productivity loss.

This result illustrates the central tradeoff. Informational and gatekeeping institutions can make default enactment much safer, but if they are too strong they erase most of the productivity gain that made default enactment attractive.

7.3 Proposal Costs

Proposal-cost screening reduces institutional load but does not automatically improve legitimacy. Across v2, default pass plus proposal costs averages 47.784 enacted bills per run and 50.195 floor bills per run, versus 81.818 and 97.500 under open default enactment. It reduces average policy shift from 0.688 to 0.607.

However, low-support passage rises from 0.455 to 0.491, and proposer gain rises from 0.686 to 1.124. This is not a software bug; it is a substantive mechanism result. If a cost function can be cleared by proposer private gain or favorable lobby pressure, costs may screen out low-value proposals while preserving or amplifying agenda-setter advantage. Proposal costs are therefore an institutional design variable, not a solved safeguard.

7.4 Challenge Vouchers

Challenge vouchers are a promising middle path. Default pass plus challenge vouchers averages 0.670 productivity, 77.968 enacted bills per run, 0.509 welfare, 0.394 low-support passage, 0.455 policy shift, 0.539 proposer gain, and 0.398 challenge use. Relative to open default enactment, it reduces productivity by 0.171, low-support passage by 0.057, policy shift by 0.208, and proposer gain by 0.117.

Table 2. Selected challenge-voucher deltas relative to open default enactment.

Case	Prod.	Low-sup.	Shift	Challenge
Baseline	-0.262	-0.065	-0.305	0.500
Low polarization	-0.182	-0.136	-0.142	0.499
High polarization	-0.294	-0.025	-0.361	0.500
Multi-party system	-0.528	-0.290	-0.559	0.815
High proposal pressure	0.018	0.014	-0.037	0.167
Extreme proposal pressure	0.077	0.024	0.022	0.100

Table 2 summarizes selected deltas from the earlier challenge sweep. The important failure mode is visible under proposal flooding: fixed token budgets are diluted as proposal volume rises.

The sweep shows that challenge design creates a throughput/safety frontier. Three party-held tokens per party increases average productivity to 0.893 but leaves low-support passage slightly worse than open default enactment. Twenty-five party-held tokens reduce low-support passage by 0.263 and policy shift by 0.478, but productivity falls by 0.454. Member-held tokens and tokenless q-member escalation reduce risk even more, but begin to resemble broad active-vote review. The lesson is not that one voucher design is optimal. The lesson is that challenge rights, token allocation, and escalation thresholds are first-order institutional parameters.

7.5 Cross-Bloc Cosponsorship

Cross-bloc cosponsorship tests agenda access before a bill reaches the default lane. In the default cross-bloc scenario, a bill must attract credible support from at least two legislators outside the proposer's party bloc and at a modest ideological distance. This gate averages 0.253 productivity, 24.696 enacted bills per run, 0.687 welfare, 0.245 low-support passage, 0.080 policy shift, and 0.236 proposer gain.

Relative to open default enactment, the gate reduces floor consideration by 0.744, low-support passage by 0.206, policy shift by 0.583, and proposer gain by 0.420. Earlier cross-bloc sweeps show that stronger gates and cross-bloc-plus-challenge combinations can reduce low-support passage further, but at large productivity cost.

This is an important negative result as much as a positive one. Coalition breadth screens for higher-welfare, lower-volatility bills, but a uniform cosponsorship gate can become too blunt. That motivates adaptive procedural tracks: routine, low-risk bills should not necessarily face the same agenda burden as high-salience, high-uncertainty, or lobby-heavy bills.

7.6 Adaptive Procedural Tracks

Adaptive procedural tracks route bills by a generated risk score rather than applying one admission rule to all proposals. The non-challenge adaptive scenario sends low-risk bills to default enactment, middle-risk bills to active simple-majority review, and high-risk bills to informed guardrails. It averages 0.442 productivity, 44.887 enacted bills per run, 0.605 welfare, 0.264 low-support passage, 0.216 policy shift, and 0.393 proposer gain.

Relative to open default enactment, that reduces productivity by 0.399 but reduces low-support passage by 0.187, policy shift by 0.447, and proposer gain by 0.262. It keeps floor consideration at 0.925, much higher than cross-bloc cosponsorship or informed guardrails. Earlier challenge-middle-lane variants preserve more productivity, but leave higher low-support passage and policy shift. The result is a useful design contrast: challenge rights preserve throughput, while active middle-lane voting removes more weakly supported bills.

7.7 Sunset Trial Review

Sunset trial review tests reversibility rather than agenda exclusion. The basic sunset-trial scenario keeps floor consideration at 1.000 because bills still enter the default-pass lane, but risky enactments that fail review expire and roll back the status quo. It averages 0.469 productivity, 47.595 enacted bills per run, 0.618 welfare, 0.374 low-support passage, 0.265 policy shift, and 0.460 proposer gain.

Relative to open default enactment, sunset review reduces productivity by 0.372 but increases welfare by 0.122, reduces policy shift by 0.398, and reduces proposer gain by 0.196. The result supports the intuition that default enactment becomes safer when the system can undo risky laws, but it also shows that reversibility is not free. In this implementation, expiration counts as failed permanent enactment.

7.8 Earned Proposal Credits

Earned proposal credits are a milder agenda-control mechanism. The base earned-credit scenario averages 0.674 productivity, 64.552 enacted bills per run, 76.252 floor bills per run, 0.513 welfare, 0.442 low-support passage, 0.508 policy shift, and 0.624 proposer gain. It denies 0.203 of potential proposals.

Relative to open default enactment, earned credits reduce productivity by 0.167, floor consideration by 0.203, low-support passage by 0.009, policy shift by 0.155, and proposer gain by 0.031, while increasing welfare by 0.017. This is not a dramatic legitimacy gain, but it is a useful contrast with static proposal costs: the mechanism reduces load without the large proposer-gain increase observed under the earlier cost formula.

Earlier combined earned-credit-plus-challenge runs were more consequential, reducing policy shift and proposer gain much more than the base credit rule alone. The result suggests that proposer accountability works best as one layer in a package, not as a substitute for contestation rights.

7.9 Budgeted Lobbying and Anti-Capture

Budgeted lobbying alone closely tracks open default enactment in the current v10 campaign: productivity remains 0.842 and enacted welfare remains 0.496, while explicit lobby spend averages 0.099 per potential bill and defensive spend accounts for 0.655 of lobbying spend. This is useful diagnostically even when the headline outcomes barely move. The simulator now distinguishes baseline lobby susceptibility from explicit organized-interest spending, including spending aimed at defeating anti-lobbying reform.

The anti-capture bundle changes outcomes more visibly. Relative to open default enactment, it changes productivity by -0.092, increases welfare by 0.022, reduces lobby capture by 0.090, and reduces private-gain ratio by 0.195. It does not solve capture outright, but it creates a measurable stress test for transparency, public-interest screening, audit sanctions, and lower direct lobby influence.

7.10 Structured Mediation

Structured mediation adds content revision before voting. In the default-pass mediation scenario, 0.906 of potential bills are amended. Productivity rises from 0.842 to 0.903, policy shift falls from 0.663 to 0.483, and proposer gain falls from 0.656 to 0.492. Compromise increases by 0.033 relative to open default-pass, while welfare changes only slightly.

This result should not be overread as mediation being universally beneficial. Under simple majority, mediation increases productivity but lowers enacted welfare and increases policy movement relative to ordinary simple majority. The more useful finding is conditional: in default-pass settings, a bounded amendment stage can preserve throughput

while reducing proposer advantage and policy volatility. That suggests future models should evaluate mediation against richer public-will and affected-group metrics rather than only chamber support.

8 Discussion

The results support the broader design premise: a legislative system cannot be judged by passage thresholds alone. Productivity, compromise, public responsiveness, proposer advantage, and minority protection are shaped by the full mechanism bundle. Open default enactment maximizes throughput in the current campaign, but it also increases low-support enactment and proposer advantage. Heavy guardrails reduce those harms but can produce a system less productive than conventional supermajority rule. Proposal costs reduce floor load, but the shape of the cost function determines which proposals survive.

Challenge vouchers illustrate a different design logic. Instead of reviewing every proposal, the institution allocates scarce contestation rights. This preserves a default lane for routine bills while forcing some risky bills into active consent. The v3 through v5 sweeps show that fixed scarce rights can be too scarce under flooding, while broad member-held or tokenless challenge rights can overcorrect into low-productivity review.

Cross-bloc cosponsorship improves proposal access upstream, before a challenge window is needed. Its limitation is that it treats coalition breadth as a universal admission requirement. That is attractive for risky or divisive bills, but wasteful for routine bills with high support and low downside. Adaptive routing responds by making procedure conditional on risk. Sunset trial review responds from the other side: it lets risky bills pass provisionally, but only renews them if they survive automatic review. Earned proposal credits add proposer learning, but only modestly change outcomes on their own. Structured mediation begins to address bargaining over bill content, but the current version still judges compromise mostly by chamber behavior. The next missing layer is richer public-will and affected-group structure so that a mediated bill can be evaluated against constituencies, concentrated harms, and representative legitimacy.

The broader contribution is methodological. Institutional design proposals often mix multiple mechanisms, making it difficult to know whether outcomes come from thresholds, agenda access, information, bargaining, lobbying, or reversibility. A modular simulator can isolate these components, compare distributions across shared worlds, and identify parameter regions where a design succeeds or fails. The aim is not to crown a winner from the current v10 scenarios; it is to create an expandable search process for discovering and stress-testing more varied legislative frameworks.

9 Limitations

This is not a validated model of the United States Congress or any real legislature. It is a mechanism-exploration simulator. The current model uses one policy dimension, generated legislators, stylized public support, stylized public benefit, simplified lobbying, no elections, one abstract mediation stage rather than full amendment bargaining, no topic-specific policy domains, and limited bicameral asymmetry.

The largest limitation is endogeneity. Real proposers choose bills strategically based on expected institutional response. The current model mostly generates proposals around proposer ideal points. Default-enactment regimes especially require endogenous proposal generation because the attractiveness of flooding, moderation, and strategic timing depends on the rules.

The second limitation is calibration. Before the simulator can support stronger empirical claims, it should reproduce rough patterns for ordinary legislative systems: coalition sizes, party unity, roll-call ideology distributions, bill progression rates, veto frequencies, and topic-specific throughput.

The third limitation is welfare specification. Public benefit is generated inside the model rather than estimated from external outcomes. It is useful for comparing institutions under common assumptions, but it should not be treated as a real social-welfare measure.

10 Research Roadmap

The next simulations should extend agenda mechanics before adding media, elections, or other realism layers. The immediate roadmap is:

(1) add richer constituent, public-will, and affected-group models so compromise is evaluated against represented people rather than only chamber support;
(2) add an active law registry with delayed sunset review, renewal, amendment, repeal, and instability metrics;
(3) add explicit route-rate metrics and threshold sweeps for adaptive tracks;
(4) add explicit cosponsor lists, reciprocity memory, and fake-cosponsorship trading to stress-test the cross-bloc gate;
(5) extend earned credits into refundable public-benefit bonds and party-level or coalition-level credit pools;
(6) deepen mediation into multi-round amendment bargaining with compensation offers for concentrated harms.

These additions directly address the observed failure modes: proposal flooding, low-support enactment, proposer gain, weak compromise formation, and permanent policy movement after weakly supported bills.

11 Conclusion

The current simulations are an early pass through a much larger design space. Default enactment is useful as a stress test because it changes the burden of collective action, but it is only one candidate. Guardrails can reduce low-support enactment, but at large productivity cost. Proposal costs reduce institutional load, but their design determines which bills survive. Challenge vouchers preserve more throughput while reducing some risk metrics, but they become diluted under flooding unless token budgets and allocation rules are tuned. Cross-bloc cosponsorship improves enacted-bill quality and reduces proposer advantage, but can become a blunt gate. Adaptive routing is a more flexible middle path. Sunset trial review shows that reversibility can reduce volatility without a front-end gate. Earned proposal credits add stateful proposer accountability and reduce policy shift modestly. Structured mediation shows that content revision can reduce proposer gain and policy movement in default-pass settings, but the model still needs richer constituent structure, affected-group harm, law registries, and multi-round bargaining.

The paper's core claim is therefore exploratory: computational institutional search can help identify legislative frameworks that better balance compromise, productivity, and representative responsiveness than any single familiar rule or isolated reform proposal.

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